

Exercise 7: Financial Forecasting

CODE:

```
public class FinancialForecast {

    public static double forecast(double presentValue, double growthRate, int years) {
        if (years == 0) {
            return presentValue;
        }
        return forecast(presentValue, growthRate, years - 1) * (1 + growthRate);
    }

    public static void main(String[] args) {
        double presentValue = 100000;
        double growthRate = 0.08;
        int years = 5;

        double futureValue = forecast(presentValue, growthRate, years);

        System.out.println("Present Value : $" + presentValue);
        System.out.println("Growth Rate   : " + (growthRate * 100) + "%");
        System.out.println("Years       : " + years);
        System.out.printf("Future Value : $%.2f%n", futureValue);
    }
}
```

OUTPUT:

```
PS D:\Github Projects\Cognizant-Hands-On-Exercise\Module 2\Exercise 2> cd "d:\Github Projects\Cognizant-Hands-On-Exercise\Module 2\Exercise 7\" ; if ($?) { javac FinancialForecast.java } ; if ($?) { java FinancialForecast }
Present Value : $100000.0
Growth Rate   : 8.0%
Years         : 5
Future Value  : $146932.81
PS D:\Github Projects\Cognizant-Hands-On-Exercise\Module 2\Exercise 7>
```